

BRANDON MANLEY

191 West Woodwuff Avenue, M2041, Columbus, OH 43210
📞 734-865-0452 ✉ manley.329@osu.edu 🌐 brandonmanley.github.io

EDUCATION

The Ohio State University

Aug. 2021 – Present

Ph.D. in Physics

Columbus, Ohio, USA

- Advisor: Prof. Yuri Kovchegov

M.S. in Physics

Aug. 2021 – Aug. 2023

- Thesis: Soft and Collinear Interactions in an Effective Field Theory

B.S. in Physics, Magna Cum Laude

Aug. 2017 – May 2021

- Honors Thesis Advisor: Prof. Stuart Raby
- Honors Thesis: Symmetries in Physical Theories

RESEARCH INTERESTS AND EXPERTISE

- Quantum chromodynamics at high energy, small- x evolution
- Spin physics: proton spin puzzle, orbital angular momentum, helicity parton distribution functions
- Phenomenology for the future Electron Ion Collider

PUBLICATIONS

1. *Elastic dijet production in electron scattering on a longitudinally polarized proton at small x : A portal to orbital angular momentum distributions*
Yuri V. Kovchegov, **Brandon Manley**
[Phys. Rev. D 111 \(2025\) 5, 054017](#)
2. *Orbital angular momentum small- x evolution: exact results in the large- N_c limit*
Brandon Manley
[JHEP 04 \(2024\) 055](#)
3. *Orbital angular momentum at small x revisited*
Yuri V. Kovchegov, **Brandon Manley**
[JHEP 02 \(2024\) 060](#)
4. *Sensitivity to millicharged particles in future proton-proton collisions at the LHC with the milliQan detector*
A. Ball et al. (incl. **Brandon Manley**)
[Phys. Rev. D 104 \(2021\) 3, 032002](#)
5. *Search for millicharged particles in proton-proton collisions at $\sqrt{s} = 13$ TeV*
A. Ball et al. (incl. **Brandon Manley**)
[Phys. Rev. D 102 \(2020\) 3, 032002](#)

In preparation:

1. *Determination of the OAM distributions from elastic dijets at the EIC: A feasibility study*
D. Adamiak et al. (incl. **Brandon Manley**)
2. *Bridging small and moderate x through exclusive π^0 production*
Shohini Bhattacharya, David DeAngelo, **Brandon Manley**

PROCEEDINGS

1. *Orbital Angular Momentum at Small x Revisited*

Brandon Manley

Contribution to SPIN 2023 proceedings

[DOI:doi.org/10.22323/1.456.0183](https://doi.org/10.22323/1.456.0183) (2024)

2. *Orbital Angular Momentum at Small x Revisited*

Brandon Manley

Contribution to DIS 2023 proceedings

[arXiv:2307.09544](https://arxiv.org/abs/2307.09544) (2023)

PRESENTATIONS

Invited talks:

- **Sep. 2025: QCD@LHC 2025**, Stony Brook University, Stony Brook, NY, USA
Feasibility of probing parton orbital angular momentum through elastic dijet production in electron-proton collisions
- **Sep. 2025: PAN seminar**, University of Connecticut, Storrs, CT, USA
Dijet production as a probe for the orbital angular momenta of partons in the proton
- **Jun. 2025: SURGE collaboration meeting**, University of California, Los Angeles, CA, USA
Probing orbital angular momentum through elastic dijet production
- **Sep. 2020: Boston University symposium**, Boston, MA, USA
Search for millicharged particles with the milliQan detector

Contributed talks and posters:

- **Feb. 2025: POETIC XI**, Florida International University, Miami, FL, USA
Poster: Elastic dijet production as a probe for orbital angular momentum distributions
- **May 2024: DeMartini poster competition**, Ohio State University, OH, USA
Poster: Orbital angular momentum at small x
- **Mar. 2024: Hayes Research Forum**, Ohio State University, OH, USA
Talk: Proton structure at high energies
- **Sep. 2023: SPIN 2023**, Durham, NC, USA
Talk: Orbital angular momentum at small x revisited
- **Jun. 2023: SURGE collaboration meeting**, Brookhaven National Lab, NY, USA
Talk: Updates on OAM at small x
- **Mar. 2023: DIS 2023**, Lansing, MI, USA
Talk: Orbital angular momentum at small x revisited
- **Mar. 2022: Hampton University Graduate Summer Program**, Jefferson Lab, VA, USA
Talk: Orbital angular momentum at small x
- **Apr. 2020: Spring Undergraduate Research Festival**, Ohio State University, OH, USA
Poster: [Machine learning approach to solve the chaotic N-body problem](#)
- **Aug. 2020: Autumn Undergraduate Research Festival**, Ohio State University, OH, USA
Talk: Search for millicharged particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

HONORS AND AWARDS

- **2024 - 2025: Presidential Fellowship**, OSU, Columbus, OH, USA
- **2024: First place, DeMartini poster competition**, OSU, Columbus, OH, USA
- **2024: Third place in Hayes Research Forum**, OSU, Columbus, OH, USA
- **2020: Summer Research Scholarship**, OSU, Columbus, OH, USA
- **2019: Summer Research Scholarship**, OSU, Columbus, OH, USA
- **2017: Maximus Merit Scholarship**, OSU, Columbus, OH, USA
- **2016: Believe in Ohio Scholarship**, Columbus, OH, USA

RESEARCH EXPERIENCE

The Ohio State University **Aug. 2021 – Present**
Graduate Research Assistant
Columbus, OH, USA

Advisor: Prof. Yuri Kovchegov (Department of Physics, OSU)

- Small x contribution of the OAM distributions to the proton spin rule
- Small x evolution of polarized dipole amplitudes and their moments
- Exclusive processes that probe the helicity PDFs and orbital angular momentum distributions at small x

The Ohio State University **Aug. 2017 – May 2021**
Undergraduate Research Assistant
Columbus, OH, USA

Advisor: Prof. Chris Hill (Department of Physics, OSU)

- Data analysis for the milliQan experiment in the search for millicharged particles
- Detector design and benchmark testing for the milliQan experiment
- GUI for the next generation pixel detector for the CMS experiment

The Ohio State University **Aug. 2020 – May 2021**
Undergraduate Research Assistant
Columbus, OH, USA

Advisor: Prof. Stuart Raby (Department of Physics, OSU)

- General symmetry structure of gauge theories and their application to isospin symmetry breaking

Kent State University **May 2016 – Aug. 2016**
Research Assistant
Kent, OH, USA

Advisor: Prof. Oleg Lavrentovich (Department of Physics, KSU)

- Structures and anisotropies of lyotropic chromonic liquid crystals

TEACHING EXPERIENCE

Department of Physics, OSU **Aug. 2021 – Present**
Graduate Teaching Assistant
Columbus, OH, USA

- Introductory Physics – Electromagnetism, Optics, Modern Physics (Head TA)
- Intermediate Classical Mechanics

Department of Mathematics, OSU **Aug. 2019 – May 2021**
Undergraduate Teaching Assistant
Columbus, OH, USA

- Precalculus, Trigonometry, College Algebra

Student Athlete Support Services Office, OSU **Aug. 2018 – May 2019**
Tutor
Columbus, OH, USA

- Calculus I, II, and III
- Introductory Physics – Electromagnetism, Optics, Modern Physics
- Intermediate Classical Mechanics

SERVICE

Society of Physics Students

Volunteer for Adopt-a-Physicist program

Oct. 2023 – Present

Columbus, OH, USA

Ohio STEP (formerly Believe in Ohio)

Volunteer Judge

Apr. 2023 – Present

Columbus, OH, USA

TECHNICAL SKILLS

Programming Languages: Python, C++, WolframScript, Julia

Technologies/Frameworks: Mathematica, ROOT, GitHub, tensorflow